SYNCHRONOUS FIREWORKS

Mithun Mohandas
readersletters@thinkdigit.com

Come October and Diwali is here! Bursting firecrackers is one of the things that come to mind. The fear that someone close to heart could get hurt is ever present when firecrackers come into the picture. So, we at Digit decided to come up with a safe way to set off firecrackers from a distance

Instructions:

The circuit relies on a simple timer IC 555 set in monostable mode to generate a pulse long enough to trigger the fireworks but not long enough to melt the heating filament.

In Fig.1, we see a resistor and a capacitor labelled R and C, respectively. It is by varying the value of these two components that we get the time period for the heating element to stay on.

Item	Value	Quantity
IC 555	NA	1
Resistor	10kΩ	2
Capacitor (Ceramic)	0.01uF	1
Capacitor (Electrolytic)	100uF 25V	1
Dry Cell	9V	1
Nichrome/ Tungsten wire	NA	1 foot
PCB	5cm X 5cm	1
Switching Transistor	2N3055	1
Connecting wires	NA	As required
Lead Acid Battery	12V 7Ah	1

Formula:

1.1 x R x C = T

R=Resistance (Ohms)
C=Capacitance (Farads)
T=Time Period (seconds)

When we used a 10kΩ (10.28kΩ on multimeter) resistor and a 100uF capacitor, our time period came down to 2.2 seconds. Theoretically, our value should've been 1.1 seconds. This can be due to varying tolerance of the components. We were able to ignite fireworks with a pulse of 1.8 seconds, so a 2.2-second pulse is capable of heating multiple heating elements. For varying the time period, use varying capacitance values and solve for R. Ensure that R is within 1kΩ - 1 1MΩ. If it's not, then vary the capacitor till you obtain R within the the permitted values.

The pulse generated from the IC555 is then used to trigger a switching transistor (2N3055) which is connected to the power source. We salvaged a 12volt 7Ah lead acid battery from an old UPS for setting off multiple ignition points. If you're using any other power source ensure that the voltage rating is well within the transistor's capabilities. Also, if you wish to use the lead acid battery, then an enclosure must be made for the entire circuit and it should be well reinforced. The enclosure must be able to withstand whatever firecracker you're going to set off. We're basically short-circuiting the battery terminals which leads to production of Hydrogen gas from the electrolyte inside. Hydrogen being highly flammable will burn at 4 per cent con-

We place the components onto the PCB prior to soldering to get a good compact arrangement. Try to avoid having any wires touch the metal transistor as it will get really hot during operation. You may extend the connecting wires of the push button to a distance safe enough for bursting firecrackers